Authentication Log Analysis and Incident Triage Lab

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Environment: Splunk Enterprise

Data: Synthetic authentication logs created for practice

Time zone: All event timestamps in this report use UTC (indicated by "Z"). Splunk's interface may display the equivalent time in IST (UTC+05:30).

Objective

Analyse login records, identify repeated authentication failures, and check whether a successful login followed those failures.

Method

Imported a CSV file into Splunk, confirmed the record count, counted failed logins by account and source IP, and arranged the selected account's events in chronological order.

Findings

The dataset contained 12 authentication events: eight failures and four successes.

Failed login counts were:

- admin01 from 203.0.113.50: five failures.
- support01 from 192.0.2.20: two failures.
- analyst03 from 192.0.2.40: one failure.

For admin01, five failures occurred between 10:00:00 and 10:02:00 UTC. A successful login followed at 10:03:00 UTC from the same source IP.

Assessment and limitations

This pattern could indicate password guessing followed by successful access. It could also result from a legitimate user entering an incorrect password repeatedly.

The logs do not include MFA results, device details, account permissions, or activity after login. Therefore, account compromise cannot be confirmed. The name admin01 alone does not establish that the account has administrator privileges.

Provisional severity

Medium for this practice scenario because repeated failures were followed by success within three minutes. This is an investigation priority, not a confirmed compromise; an organisation's severity policy and additional evidence could change the rating.

Recommended escalation

Escalate to a senior SOC analyst for further investigation. Recommend checking:

- Whether the account owner recognises the login attempts.
- Whether the source IP and device are expected.
- MFA results and the account's actual permissions.
- Actions performed after the successful login.

Evidence

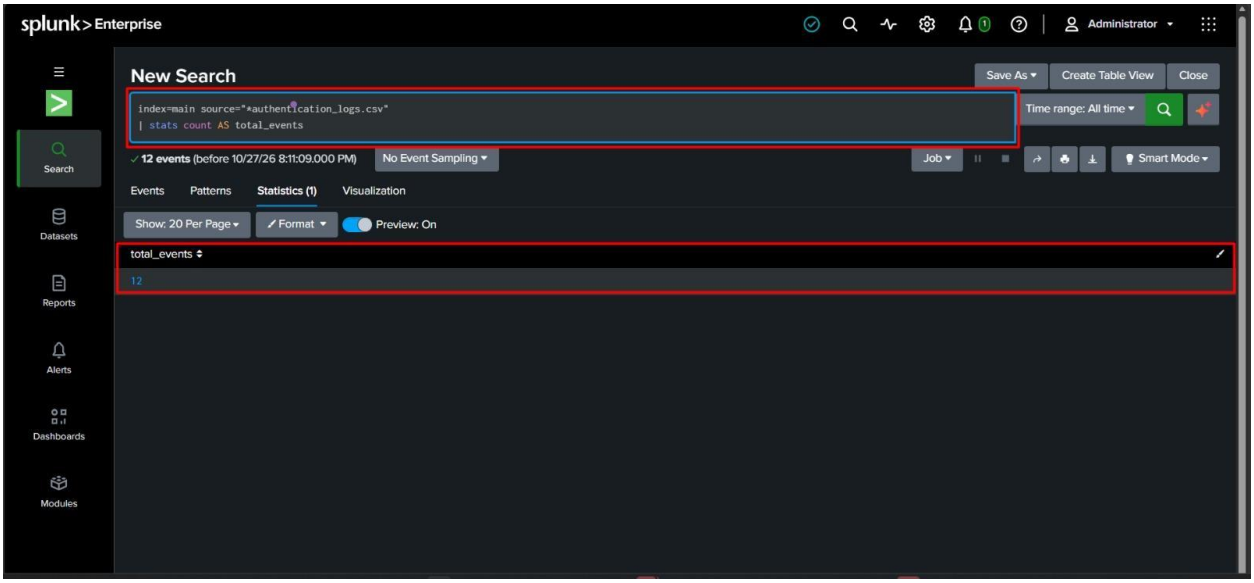


Figure 1: Total authentication events imported into Splunk (01_total_events.png)



Figure 2: Failed login counts by account and source IP (02_failed_login_counts.png)

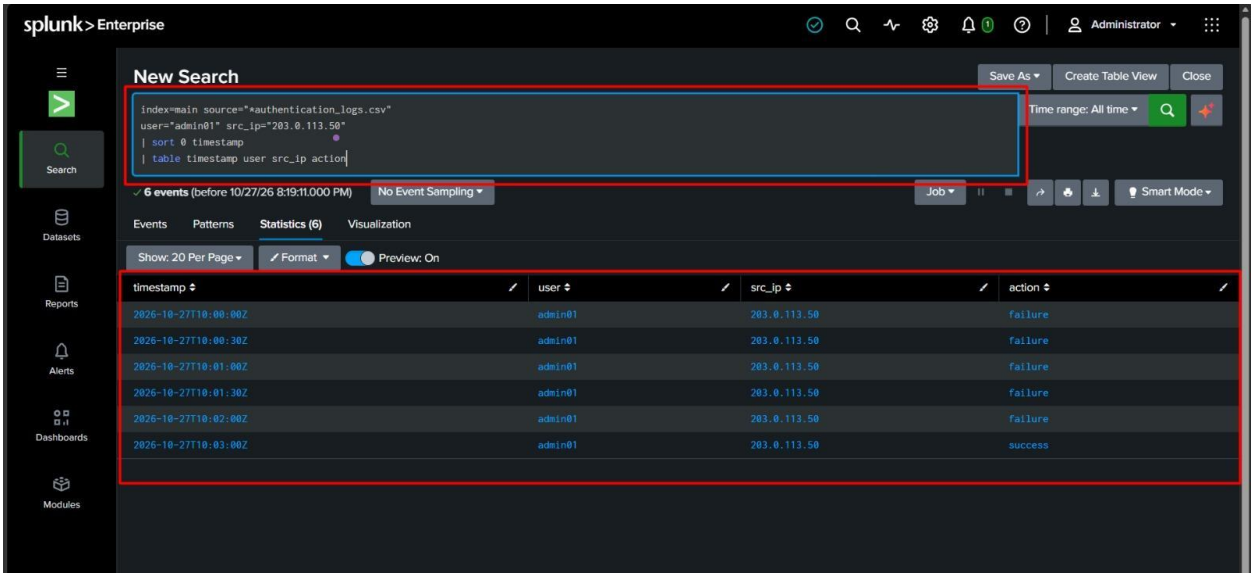


Figure 3: Five failed logins followed by success for admin01 (03_admin_login_timeline.png)

Conclusion

Identified repeated failed logins followed by a successful login and documented the evidence. Further investigation is required to determine whether the activity was legitimate or malicious.